Long Nguyen Khac

Hanoi, Vietnam | nk.long723@gmail.com | longnk0702.github.io | longnk0702 | longnk0702

Education

International School, Vietnam National University, Hanoi — *Vietnam*
B.Eng. in Automation and Informatics — *Sep. 2021 – Dec. 2026*

- **GPA:** 3.71/4.00; ranked in the top 4% of the cohort.
- **Gap year:** exchanged in Taiwan (CCU) and Singapore (NUS & NTU). (05/2025 – 05/2026).
- **Key Objective:** to graduate with Excellent distinction with the highest academic standing in the cohort.
- **Research interest:** focusing on adaptive control for AUVs integrated with wireless power charging system.

Research Experience

Water & Energy Research Lab, EEE, Nanyang Technological University — *Singapore*
Research Fellow — *Feb. 2026 – Mar. 2026*

- Supervisor: Asst. Prof. Yun Yang & PhD. Kaiyuan Wang
- Project: Advanced 3D Wireless Power Transfer System for AUVs

AISC Lab, AIM-HI, National Chung Cheng University — *Taiwan / Remote*
Research Intern — *May 2025 – Present*

- Supervisor: Prof. Her-Terng Yau & PhD.Hao-Yang Lin
- Project: Prediction of Cutting Force and Tool Wear Using Machine Learning Methods

Intelligent Control and Artificial Intelligence Laboratory, VNUIS — *Vietnam*
Student Researcher — *Sep. 2022 – Present*

- Supervisor: Dr. Le Xuan Hai
- Project 1: Adaptive Control for AUV with Wireless Power Transfer System
- Project 2: Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Delta Robot
- Project 3: Prediction of Personal Protective Equipment Using YOLOv5s

International Experience

Nanyang Technological University — *Singapore*
Global Connect Fellow — *Feb. 2026 – Mar. 2026*

- **Program:** NTU Global Connect Fellowship
- Selected as 1 of 39 young scholars worldwide and 1 of 5 outstanding Vietnamese young scholars funded fully for the Global Connect Fellowship program at NTU, Singapore (∼2% acceptance rate from ∼2,700 applicants).

National University of Singapore — *Singapore*
Exchange Student — *Aug. 2025 – Dec. 2025*

- **Program:** DiscoverNUS
- Selected as one of 8 outstanding VNU students receiving a fully funded scholarship DiscoverNUS & the NUS ASEAN Master's Scholarship Programme.

National Chung Cheng University — *Taiwan*
Research Intern — *May. 2025 – Jul. 2025*

- **Program:** Taiwan Experience Education Program (TEEP)
- Selected as one of 28 global students to intern at AIM-HI, CCU funded by Ministry of Education, Taiwan

Universitas Gadjah Mada & Kyushu University — *Indonesia*
VNU representative — *Feb. 2025*

- **Program:** ASEAN in Today's World
- Selected as one of four excellent VNU representatives to participate the program funded by Kyushu University, Japan.

Ngee Ann Polytechnic — *Singapore*
VNU Representative — *Feb. 2024*

- **Program:** Temasek Foundation Specialist's Community Action & Leadership Exchange
- Selected as one of 29 VNU students to receive a fully-funded scholarship for the program by Temasek Foundation.

Projects

Adaptive Control for AUV with Wireless Power Transfer System (Graduation Thesis) — *Vietnam*
Leader — *Apr. 2026 – Present*

- Developing an AUV prototype that integrates motion control with wireless charging requirements.
- Focusing on station keeping and stable positioning during planar wireless power transfer.

Advanced 3D Wireless Power Transfer System for AUVs *Singapore*
Leader *Feb. 2026 – Mar. 2026*

- Studied relay-coil-enhanced spatial wireless power transfer systems for autonomous underwater vehicle charging.
- Analyzed coil configurations, equivalent circuit behavior, power transfer efficiency, and load-dependent system performance.
- Got Most Popular Poster Award in NTU GCF Final Research Presentation

Prediction of Cutting Force and Tool Wear Using Machine Learning Methods *Taiwan/Remote*
Leader *May. 2025 – Present*

- Applying machine learning models (Random Forest, XGBoost, and Support Vector Regression) to Taguchi machining datasét.
- Built data-driven models for cutting force and tool wear prediction in CNC milling using experimental machining datasets.

Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Delta Robot *Vietnam*
Member *Oct. 2023 – Jul. 2024*

- Designed and built a 3-DOF delta parallel robot prototype for motion control research.
- Implemented adaptive fuzzy backstepping control to improve trajectory tracking under uncertainties.
- Presented in International Conference on Advances in Information and Communication Technology 2024.

Prediction of Personal Protective Equipment Using YOLOv5s *Vietnam*
Member *Jan. 2023 – Jul. 2023*

- Trained YOLOv5s-based object detection models and compared six activation functions for performance improvement.
- Evaluated model performance using precision, recall, mAP@0.5, and mAP@0.5–0.95.
- Presented in National Conference on Smart Technology Application in Industry 4.0, Smart City, and Sustainability 2023.

Publications

1. A. T. Le, G. L. Truong, H. Q. Nguyen, T. A. T. Ngoc, B. T. The, **L. N. Khac**, et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Delta Robot.* International Conference on Advances in Information and Communication Technology (ICTA), Springer, 2024. DOI: 10.1007/978-3-031-80943-9_57.
2. D. M. Tuan, T. M. T. Kiet, N. V. Tung, N. X. The, N. A. Chau, P. T. Ngoc, **N. K. Long**, et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites.* Proceedings of the National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability (STAIS), 2023.

Honors & Awards

Most Popular Poster Award, NTU Singapore Global Connect Fellowship Final Research Poster Presentation	Mar. 2026
NUS ASEAN SPARK Fellow, NUS Enterprise Summer Programme in Entrepreneurship	Jul. 2026
Outstanding Young Face Award, Vietnam National University, International School, awarded in 2024 and 2025	2024 – 2025
Five Good Merits Award, Vietnam National University, Hanoi	Dec. 2024
Toshiba Scholarship, ToshibaInternational Foundation	Sep. 2024
Fighting Spirit Prize, The IEEE Southeast Asia Circuits and Systems Society Hackathon	Dec. 2022
Merit-based University Academic Scholarship, Five-semester recipient for excellent academic performance	2021–2025

Skills

Programming	C/C++, MATLAB/Simulink, Python, LaTeX, PLC
Hardware	STM32(ARM-Cortex-M), ESP32, NVIDIA Jetson Nano, Arduino Uno
Software	MATLAB, SolidWorks, Altium, Proteus, Visual Components
Robotics	Kinematics (Forward, Inverse), Control Theory (PID, SMC, Backstepping, MPC)
Development workflow	Git, Notion, Oscilloscope, Multimeter
Languages	Vietnamese (Native), English (IELTS 6.5), French (Beginner), Chinese (Beginner)

References

Prof. Her-Terng Yau Director of AIM-HI, National Chung Cheng University
Her-Terng Yau | htyau@ccu.edu.tw

Asst. Prof. Yun Yang Assistant Professor, School of Electrical & Electronic Engineering, NTU
Yun Yang | yun.yang@ntu.edu.sg

Dr. Le Xuan Hai Vice Dean, Faculty of Engineering and Technology, VNUIS
Hai-Xuan Le | hailx@vnu.edu.vn